Adnan Ahmad

Machine Learning Researcher — Multimodal AI, LLMs & Distributed Learning

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Professional Profile

Machine Learning researcher with a PhD and 13+ peer-reviewed publications, with expertise in federated and decentralized learning, multimodal AI, and large-scale deep learning for privacy-preserving intelligent systems. Research spans representation learning, continual learning, foundation-model adaptation, and retrieval-augmented generation (RAG), with applied experience developing AI methods for multimodal healthcare data, physiological signal analysis, and real-time intelligent systems.

Research Areas

- Federated and decentralized learning for privacy-preserving AI across heterogeneous, distributed data environments.
- Foundation-model adaptation, parameter-efficient fine-tuning, large language models, and retrieval-augmented generation (RAG).
- Multimodal deep learning, representation learning, medical image analysis, and physiological signal modelling for healthcare AI.
- Continual learning, self-supervised learning, and adaptive ML under distribution shift and non-stationary environments.

Education

Doctor of Philosophy in Information Technology (Machine Learning) — **Dec 2020 – Sep 2024**
Deakin University — *Burwood, VIC, Australia*

Thesis: Robust and Communication-Efficient Federated Learning under Non-IID Data. Developed optimization and aggregation algorithms for privacy-preserving federated learning under statistically heterogeneous data, addressing communication-efficient distributed optimization, scalable deep learning, and decentralized learning via second-order optimization and representation learning on benchmark computer-vision datasets.

Bachelor of Science in Electrical Engineering (Computer Engineering) — **Sep 2013 – Sep 2017**
COMSATS University — *Islamabad, Pakistan*

Final Year Project: FPGA implementation of general-purpose processors and hardware accelerators for computer-vision applications, compared on throughput-to-area ratio.

Research Experience

Associate Research Fellow (Machine Learning & Multimodal AI) — **March 2024 – May 2025**
Deakin University — *Waurn Ponds, VIC, Australia*

- Led an AOARD-funded project developing multimodal ML methods for healthcare physiological data, integrating EEG, ECG, eye-tracking, and behavioural signals for robust human-state modelling and intelligent decision support.
- Developed deep, continual, self-supervised, and representation-learning methods that improved cross-subject generalization and robustness under distribution shift in multimodal biomedical data.
- Built real-time AI systems for multimodal data acquisition and inference using Lab Streaming Layer (LSL), Docker, and Google Cloud Platform, with low-latency pipelines for continuous physiological signal processing.
- Co-supervised PhD and Master's students, contributed to ethics applications and competitive research grants, and led research dissemination and outreach.

Machine Learning Researcher (Decentralized Federated Learning) — **July 2024 – Jan 2025**
National Research Council of Italy (IIT-CNR) — ERCIM Fellowship — *Pisa, Italy*

- Developed privacy-preserving optimization and aggregation algorithms for heterogeneous distributed AI systems.
- Designed a second-order (Hessian-informed) decentralized aggregation framework that improved convergence and communication efficiency under non-IID data, without a central parameter server.
- Implemented scalable distributed-learning frameworks in PyTorch, evaluated across heterogeneous peer-to-peer network topologies on benchmark computer-vision datasets.
- Collaborated with an international team on algorithm development, large-scale experimentation, publications, and open-source research software.

Casual Academic (Teaching Assistant — Machine Learning) — **March 2021 – March 2024**
Deakin University — *Burwood, VIC, Australia*

- Delivered workshops in deep learning, machine learning, and computational intelligence, covering computer vision, NLP, neural networks, optimization, and intelligent systems.

Publications

Full list: Google Scholar

Journal Articles

1. A. Ahmad, W. Luo, and A. Robles-Kelly, "Robust federated learning under statistical heterogeneity via Hessian-weighted aggregation," *Machine Learning*, vol. 112, 2023.
2. A. Ahmad, W. Luo, and A. Robles-Kelly, "Robust federated learning under statistical heterogeneity via Hessian spectral decomposition," *Pattern Recognition*, vol. 141, 2023.
3. A. Ahmad, B. Nakisa, and M. N. Rastgoo, "Robust emotion recognition via bi-level self-supervised continual learning," *Neurocomputing*, 2026.
4. A. Ahmad, B. Nakisa, and M. N. Rastgoo, "Adaptive prototype aggregation for continual learning on physiological data under cross-subject variability," *ACM Transactions on Intelligent Systems and Technology*, 2026.
5. S. R. Naqvi, A. Ahmad, S. M. R. Islam, et al., "Towards prevention of sportsmen burnout: formal analysis of sub-optimal tournament scheduling," *Computers, Materials & Continua*, vol. 70, 2022.
6. A. Ahmad, C. Boldrini, L. Valerio, A. Passarella, and M. Conti, "DecHW: Heterogeneous decentralized federated learning exploiting second-order information," *IEEE Transactions on Emerging Topics in Computational Intelligence*, *under review*.
7. A. Ahmad, B. Nakisa, and M. N. Rastgoo, "Partner-aware hierarchical skill discovery for robust human-AI collaboration," *Expert Systems with Applications*, *under review*.
8. A. Ahmad, B. Nakisa, and M. N. Rastgoo, "Adaptive human-AI coordination via hierarchical action disentanglement," *Transactions on Machine Learning Research*, *under review*.

Conference Proceedings

9. A. Ahmad et al., "Bio-inspired dual-network model to tackle statistical heterogeneity in federated learning," 2023.
10. A. Ahmad, W. Luo, and A. Robles-Kelly, "Federated learning under statistical heterogeneity on Riemannian manifolds," in *PAKDD*, 2023.
11. A. Ahmad, W. Luo, and A. Robles-Kelly, "A Hessian-based federated learning approach to tackle statistical heterogeneity," in *Advanced Data Mining and Applications*, Springer, 2023.
12. N. Fernando, B. Nakisa, A. Ahmad, and M. N. Rastgoo, "Adaptive XAI in high-stakes environments: modeling swift trust with multimodal feedback in human-AI teams," 2025.
13. D. Spina et al., "Report on the 3rd workshop on neurophysiological approaches for interactive information retrieval (NeurophysIIR 2025) at ACM SIGIR CHIIR 2025," 2025.

Reviewer Activities

Reviewer for NeurIPS (2026), ICLR (2025), Elsevier *Pattern Recognition*, Elsevier *Pervasive and Mobile Computing*, PAKDD, IEEE IJCNN, and the WACV Workshop on Complex-Valued Deep Learning / SARFish Challenge.

Awards & Recognition

- ERCIM "Alain Bensoussan" Fellowship Programme — Offered 2024
- Postgraduate Research Scholarship, Deakin University — 2020–2024
- Chartered Professional Engineer (CPEng), Engineers Australia
- Registered Engineer, Pakistan Engineering Council (PEC)
- Deakin Cyber Research Grant — AUD 12,000

Technical Skills

Machine Learning: Federated & Decentralized Learning, Multimodal Learning, Foundation Models, LLMs, Parameter-Efficient Fine-Tuning, Continual / Self-Supervised / Representation Learning, Reinforcement Learning
Programming: Python, SQL, C++, Java, Verilog
Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, SentenceTransformers, Scikit-learn, Pandas, NumPy, OpenCV
LLM & Retrieval: RAG, Vector Databases (Qdrant), Prompt Engineering
Cloud & MLOps: Docker, FastAPI, Git, MLflow, Weights & Biases, GCP (Compute Engine, Vertex AI), LSL
Distributed & Hardware: Peer-to-Peer Training, HPC, CUDA, NVIDIA Jetson Nano, FPGA (Xilinx Virtex-5)

Selected Projects

- **Decentralized Federated Learning with Second-Order Optimization —** PyTorch peer-to-peer FL framework with a Hessian-based aggregation method (DecHW) that outperforms FedAvg on convergence and communication cost under non-IID data. Project | GitHub
- **Real-Time Human–AI Coordination via Physiological Feedback —** Overcooked system where models infer human mental state from EEG/ECG/eye-tracking and RL agents adapt; built on LSL streaming, Docker, and FastAPI. Project | GitHub
- **Real-Time Conversational AI for Pepper Robot —** Low-latency speech-to-response system integrating a SoftBank Pepper robot with GPT via an API-driven architecture. Project | GitHub
- **Pipelined MIPS Processor & Vision Accelerators on FPGA —** Pipelined MIPS processor and low-area vision accelerators (Verilog) on Xilinx Virtex-5, plus a Java assembler, for hardware–software co-design and latency/area/throughput analysis. Project | GitHub

Outreach

Student Ambassador, Deakin University (2022–2024) — delivered AI and robotics demonstrations to large audiences during Open Day.